

Total no. of Pages:

Roll no.

Degree: B. Tech Semester: Third Sem Course work
END-SEMESTER EXAMINATION, Nov-Dec 2023
Course Title: Digital Circuits and Systems
Course Code: EAEP08

Time: 03 Hours

Max. Marks: 40

Note: - Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	Explain the need for and importance of error detecting and error correcting code. Find the position of error in the following received data using even parity error detection assuming four-bit original data: Data received = (1110101)₂	4	01
1b	Implement the following Boolean function F, together with the don't-care conditions d, using no more than two NOR gates. Assume that both the normal and complement inputs are available. F(A, B, C, D) = $\sum (2, 4, 10, 12, 14)$ D(A, B, C, D) = $\sum (10, 1, 5, 8)$	4	01
1c	i. Define the following parameters: a, Propagation Delay b, Fan-Out c, V _{IH} d, Rise time. ii. Draw the CMOS circuit for the given expression: F = ((A+B) C + A' B)'	2 2	02
Q 2	Attempt any 2 parts of the following		
2a	Design a 4-bit incrementor circuit	4	03
2b	Explain the need of carry-look ahead generator and draw a 4-bit carry-look ahead generator.	4	03
2c	i. Implement the given two functions using PLA with minimum number of product terms: F1(A, B, C) = $\sum m(0, 1, 2, 4)$ F2(A, B, C) = $\sum m(0, 5, 6, 7)$ ii. The inputs of a 4 to 1 MUX are as follows: I₀ = Z, I₁ = Z', I₂ = '0', I₃ = '1' . The select lines are connected to the inputs of the function X & Y. Z is also the input of the function. Find the function which the multiplexer implements	2 2	05 03
Q 3	Attempt any 2 parts of the following		
3a	What is the difference between ring counter and Johnson's ring counter. Draw the block diagram and timing diagram of 4-bit Johnson's ring counter and explain.	4	03

3b	Draw the Mod-10 ripple counter and explain its working with the help of timing diagram.	4	03
3c	Design a MOD-3 up-down synchronous counter using T- Flipflop.	4	05
Q 4	Attempt any 2 parts of the following		
4a	Explain the importance of digital circuit simulation using VHDL and explain the hardware abstraction using entity-architecture pair.	4	04
4b	Write the VHDL code of half adder using behavioural and structural modelling.	4	04
4c	Write the VHDL code for 4-bit SISO shift register.	4	04
Q 5	Attempt any 2 parts of the following		
5a	Design a serial adder using JK Flipflop.	4	05
5b	Draw the state diagram to detect the sequence "1101" using mealy model and implement using D-Flipflop.	4	05
5c	Write notes on the following: i.CPLD ii.FPGA	2 2	05